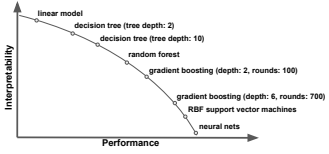
Interpretable Machine Learning

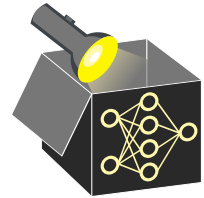
Introduction, Motivation, and History



Learning goals

- Why interpretability?
- Developments until now?
- Use cases for interpretability

WHY INTERPRETABILITY?



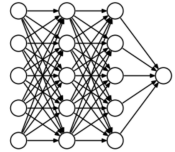
- ML: huge potential to aid decision-making process due to its predictive performance
- ML models are black boxes, e.g., XGBoost, RBF SVM or DNNs
  $\rightsquigarrow$ too complex to be understood by humans
- Some applications are "learn to understand"
- When deploying ML models, lack of explanations
  1. hurts trust
  2. creates barriers

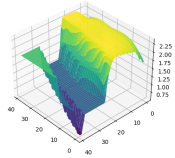
$\rightsquigarrow$ Many disciplines with required trust rely on traditional models, e.g., linear models, with less predictive performance

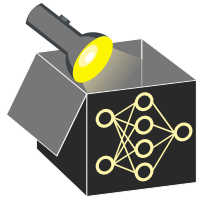
WHY INTERPRETABILITY?

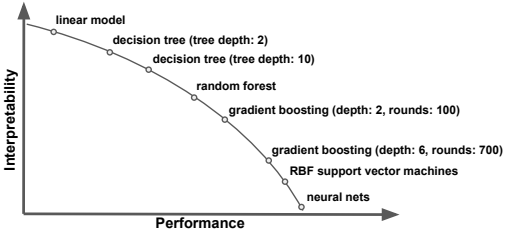
- ML: huge potential to aid decision-making process due to its predictive performance
- ML models are black boxes, e.g., XGBoost, RBF SVM or DNNs
  $\rightsquigarrow$ too complex to be understood by humans
- Some applications are "learn to understand"
- When deploying ML models, lack of explanations
  1. hurts trust
  2. creates barriers

$\rightsquigarrow$ Many disciplines with required trust rely on traditional models, e.g., linear models, with less predictive performance

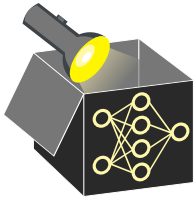






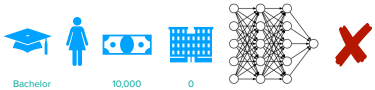


INTERPRETABILITY IN HIGH-STAKES DECISIONS

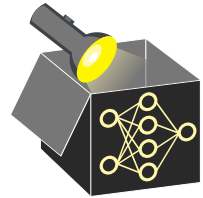


Examples of critical areas where decisions based on ML models can affect human life

- Credit scoring and insurance applications
  - Society of Actuaries
    - Reasons for not granting a loan
    - Fraud detection in insurance claims



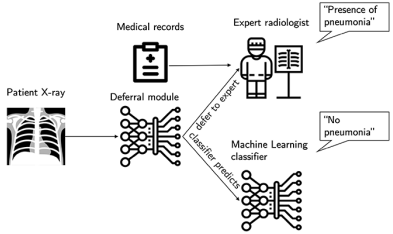
INTERPRETABILITY IN HIGH-STAKES DECISIONS



Examples of critical areas where decisions based on ML models can affect human life

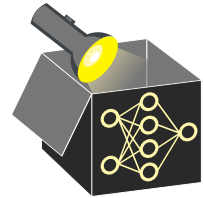
- Credit scoring and insurance applications
  ▸ Society of Actuaries
  - Reasons for not granting a loan
  - Fraud detection in insurance claims

- Medical applications
  - Identification of diseases
  - Recommendations of treatments
- ...



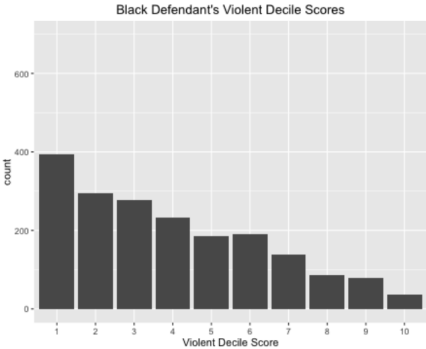
▸ Miliard (2020)

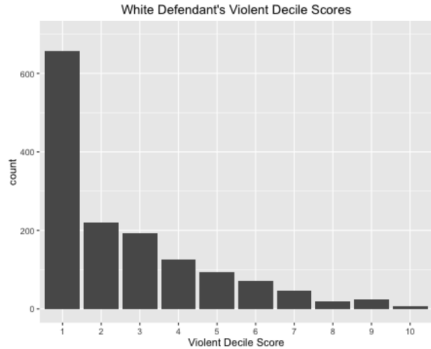
NEED FOR INTERPRETABILITY



Need for interpretability becoming increasingly important from a legal perspective

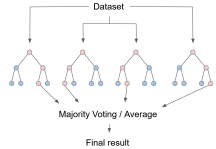
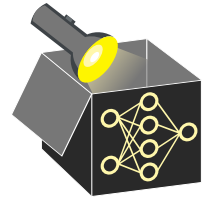
- General Data Protection Regulation (GDPR) requires for some applications that models have to be explainable ▸ Goodman & Flaxman (2017)
  $\rightsquigarrow$ *EU Regulations on Algorithmic Decision-Making and a "Right to Explanation"*
- *Ethics guidelines for trustworthy AI* ▸ European Commission (2019)





BRIEF HISTORY OF INTERPRETABILITY

- 18th and 19th century:
  linear regression models (Gauss, Legendre, Quetelet)
- 1940s:
  emergence of sensitivity analysis (SA)
- Middle of 20th century:
  Rule-based ML, incl. decision rules and decision trees
- 2001:
  built-in feature importance measure of random forests
- >2010:
  Explainable AI (XAI) for deep learning
- >2015:
  IML as an independent field of research

▸ Carl Friedrich Gauss

▸ Wikipedia